ORF309 MIDTERM 2 CHEAT SHEET – Poisson, Random Walks, Joint Densities

1. POISSON PROCESSES

Basic Facts

$N(t) \sim \text{Poisson}(\lambda t)$

$$P(N(t)=k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$
$$E[N(t)] = \text{Var}(N(t)) = \lambda t$$

Thinning (MOST IMPORTANT!)

If arrivals $\sim$ Poisson(λ), each Type A w/ prob p:

$$\boxed{\text{Type A} \sim \text{Poisson}(\lambda p)}$$

Example: Customers at rate 2/min, 40% coffee
$\Rightarrow$ Coffee orders: Poisson($2 \times 0.4 = 0.8$/min)

Superposition

Independent Poisson(λ_1) + Poisson(λ_2)
= Poisson($\lambda_1 + \lambda_2$)

Competing Processes

Process A at rate λ_A, B at rate λ_B:

$$P(\text{A before B}) = \frac{\lambda_A}{\lambda_A + \lambda_B}$$
$$E[\text{time}] = \frac{1}{\lambda_A + \lambda_B}$$

Conditional Expectation

Given n arrivals occurred:

$$\boxed{E[T \mid N(T) = n] = \frac{n}{\lambda}}$$

Example: 10 customers at rate 2/min
$\Rightarrow E[T] = 10/2 = 5$ min

2. RANDOM WALKS

Setup

- States: $a, \ldots, i, \ldots, b$ (barriers at a, b)
- Start at i, move w/ probs $p_\uparrow, p_\downarrow, p_\leftrightarrow$

First-Step: Hitting Probability

$p(i) = P(\text{hit } b \text{ before } a \mid \text{start } i)$

Recursive:

$$p(i) = p_\uparrow p(i+1) + p_\downarrow p(i-1) + p_\leftrightarrow p(i)$$

Boundaries: $p(a) = 0$, $p(b) = 1$

Symmetric shortcut ($p_\uparrow = p_\downarrow = 1/2$):

$$\boxed{p(i) = \frac{i-a}{b-a}}$$

First-Step: Expected Time

$\mu(i) = E[\text{time to barrier} \mid \text{start } i]$

Recursive:

$$\mu(i) = \mathbf{1} + p_\uparrow \mu(i+1) + p_\downarrow \mu(i-1) + p_\leftrightarrow \mu(i)$$

Boundaries: $\mu(a) = 0$, $\mu(b) = 0$
DON'T FORGET THE +1!

Symmetric shortcut:

$$\boxed{\mu(i) = (i-a)(b-i)}$$

Problem-Solving Steps

1. Draw diagram, identify $p(i)$ or $\mu(i)$
2. Check if symmetric $\rightarrow$ use shortcut
3. Write boundaries
4. Write equations for all middle states
5. Solve system

3. JOINT/CONDITIONAL DENSITIES

Key Relationships

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$
$$f_{X,Y}(x,y) = f_{Y|X}(y|x) \cdot f_X(x)$$
$$f_X(x) = \int f_{X,Y}(x,y)\,dy$$

Law of Total Expectation

$$\boxed{E[Y] = E[E[Y|X]]}$$
$$= \int E[Y|X=x] \cdot f_X(x)\,dx$$

The $Y = UX$ Transform (Common!)

$U \sim \text{Unif}[0,1]$ indep. of X, $Y = UX$:

$$f_{Y|X}(y|x) = \frac{1}{x} \quad (0 < y < x)$$
$$E[Y|X=x] = \frac{x}{2}$$
$$f_{X,Y}(x,y) = \frac{f_X(x)}{x} \quad (0 < y < x)$$

Always specify range!

Strategy

- Need $E[Y]$? $\rightarrow$ Try $E[E[Y|X]]$
- Building joint? $\rightarrow$ $f_{X,Y} = f_{Y|X} \cdot f_X$
- Draw picture of region!

4. EXPONENTIAL

$T \sim \text{Exp}(\lambda)$:

$$f(t) = \lambda e^{-\lambda t} \quad (t > 0)$$
$$P(T > t) = e^{-\lambda t}$$
$$E[T] = \frac{1}{\lambda} \quad (\text{NOT } \lambda!)$$
$$\text{Var}(T) = \frac{1}{\lambda^2}$$

Memoryless Property

$$P(T > s + t \mid T > s) = P(T > t)$$

Min/Max of Exponentials

$T_1 \sim \text{Exp}(\lambda_1)$, $T_2 \sim \text{Exp}(\lambda_2)$ indep:

$$\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$$
$$P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$
$$E[\max(T_1, T_2)] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

5. BROWNIAN MOTION

$B(t)$ standard BM:

$$B(0) = 0$$
$$B(t) \sim N(0, t)$$
$$E[B(t)] = 0, \quad \mathrm{Var}(B(t)) = t$$
$$B(t) - B(s) \sim N(0, t-s) \quad (t > s)$$

Conditional:

$$E[B(t) \mid B(s) = x] = x$$
$$E[B(t)^2 \mid B(s) = x] = x^2 + (t - s)$$

Geometric BM: $S(t) = S_0 e^{B(t)}$ (stocks)

PATTERN RECOGNITION

See	**Use**
Types/categories	Thinning
Which first?	Competing
Given n events	$E[T] = n/\lambda$
Position/game	Random walk
Prob reaching?	$p(i)$ equations
Expected time?	$\mu(i)$ (+1!)
Two variables	Joint PDF
$Y = UX$	$f_{Y\|X} = 1/x$
Waiting time	Exponential
Min/first	Sum rates

COMMON MISTAKES

1. Forgetting +1 in $\mu(i)$ formula
2. Not using thinning for types
3. Confusing λ with $1/\lambda$
4. Wrong boundaries: $\mu(a) = \mu(b) = 0$
5. Not specifying range for $f_{X,Y}$
6. Not checking symmetry first

OTHER (BACKUP)

Bin(n, p): $E = np$, Var$= np(1-p)$ **Geom**(p): $E = 1/p$
Unif$[a, b]$: $E = \frac{a+b}{2}$
Normal: $E = \mu$, Var$= \sigma^2$, $Z \sim N(0, 1)$
Gamma(k, λ): $E = k/\lambda$, Var$= k/\lambda^2$

ORF309 MIDTERM 2 CHEAT SHEET (PAGE 2) – Worked Examples

POISSON EXAMPLES

Ex 1-3: Poisson Basics

Thinning: $2 \times 0.4 = 0.8$/min Competing: $P(\text{Uber 1st}) = \frac{6}{10}$ Conditional: $E[T|N = 10] = 5$ min

Ex 4: Thinning + Competing

Q: Coffee shop: 2/min arrive, 40% coffee, 20% food. P(4 coffee before 1st food)?

A: $\lambda_c = 0.8$, $\lambda_f = 0.4$. Among c+f only, each is c w/ prob 2/3:

$$P = \left(\frac{2}{3}\right)^4 \cdot \frac{1}{3} = \frac{16}{243}$$

RANDOM WALK EXAMPLES

Ex 5-7: Random Walk

Symm 0–4, start 2: $p(2) = \frac{2}{4}$, $\mu(2) = 2 \cdot 2 = 4$ Biased: write equations, solve system

JOINT DENSITY EXAMPLES

Ex 8: $Y = UX$ Transform

$P(X > x) = 1/x^2 \Rightarrow f_X = 2/x^3$. $E[Y|X = x] = x/2 \Rightarrow E[Y] = \frac{1}{2}E[X] = 1$

Ex 9-10: Joint/Conditional

$f_{X,Y}(x, y) = \frac{2}{x^4}$ for $0 < y < x$, $x \geq 1$ $f_Y(1) = \frac{2}{3}$ $P(X > 2|Y = 1) = \frac{1}{8}$

EXPONENTIAL EXAMPLES

Ex 11-14: Exponential

Exp(2): $E = 0.5$, $P(> 3) = e^{-6}$ Min(6,4): $E = 0.1$hr Max(1/8,1/4): $E = 8 + 4 - 8/3 = 9.33$hr

BROWNIAN MOTION EXAMPLES

Ex 15-18: BM

$\text{Var}(B(5)) = 5$ $B(5) - B(2) \sim N(0, 3)$ $E[B(5)^2|B(2) = 3] = 12$

Geom BM: $S(t) = e^{B(t)}$, $P(S(2) \geq S(1)^2) = P(B(2) - 2B(1) \geq 0) = 0.5$

QUICK REFS

$P(A|B) = \frac{P(A\cap B)}{P(B)}$ $E[aX + bY] = aE[X] + bE[Y]$ $\text{Var}(X) = E[X^2] - (E[X])^2$

Indep: $f_{X,Y} = f_X \cdot f_Y$ IBP: $\int udv = uv - \int vdu$

EXAM STRATEGY

Time: 35 min/Q, 5 min check. **Approach:** Read, identify technique, write setup, show work, check sense.

Partial credit: Write formulas, define variables, explain approach. **Check:** $P \in [0, 1]$? PDF = 1? Units?

Key: +1 in $\mu(i)$ | Thinning for types | $E[T] = 1/\lambda$ | Range for PDFs | Big 3 = 75%!